

# IBSIMU Beam Extraction Report

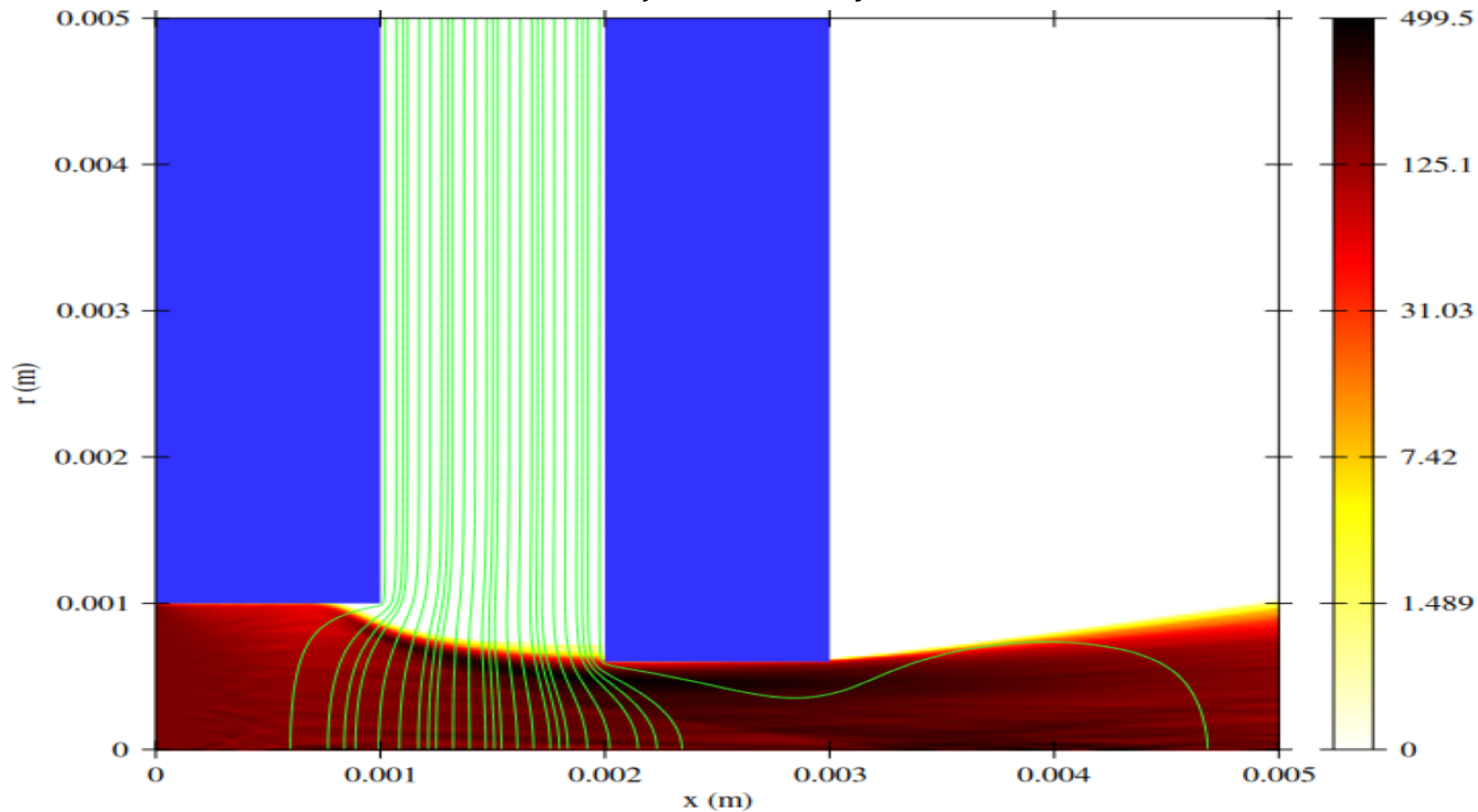
2026-09-10 18:03

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--- Simulation Parameters ---
Grid points: 241
Particles: 5000
Current density: 600 A/m²
Beam energy: 5.0 eV
Species: Ar⁺
Mass: 39.948 amu, Charge: 1.0 e
Solver: BiCGSTAB
Beam mode: Energy/Temp
B-field: None

--- Electrodes ---
#      Gap      Apt    V(kV)    Thk    Chm    Wall
1       0.0       1     0.0       1      0      0
2       1.0      0.6   -1.34      1      0      0

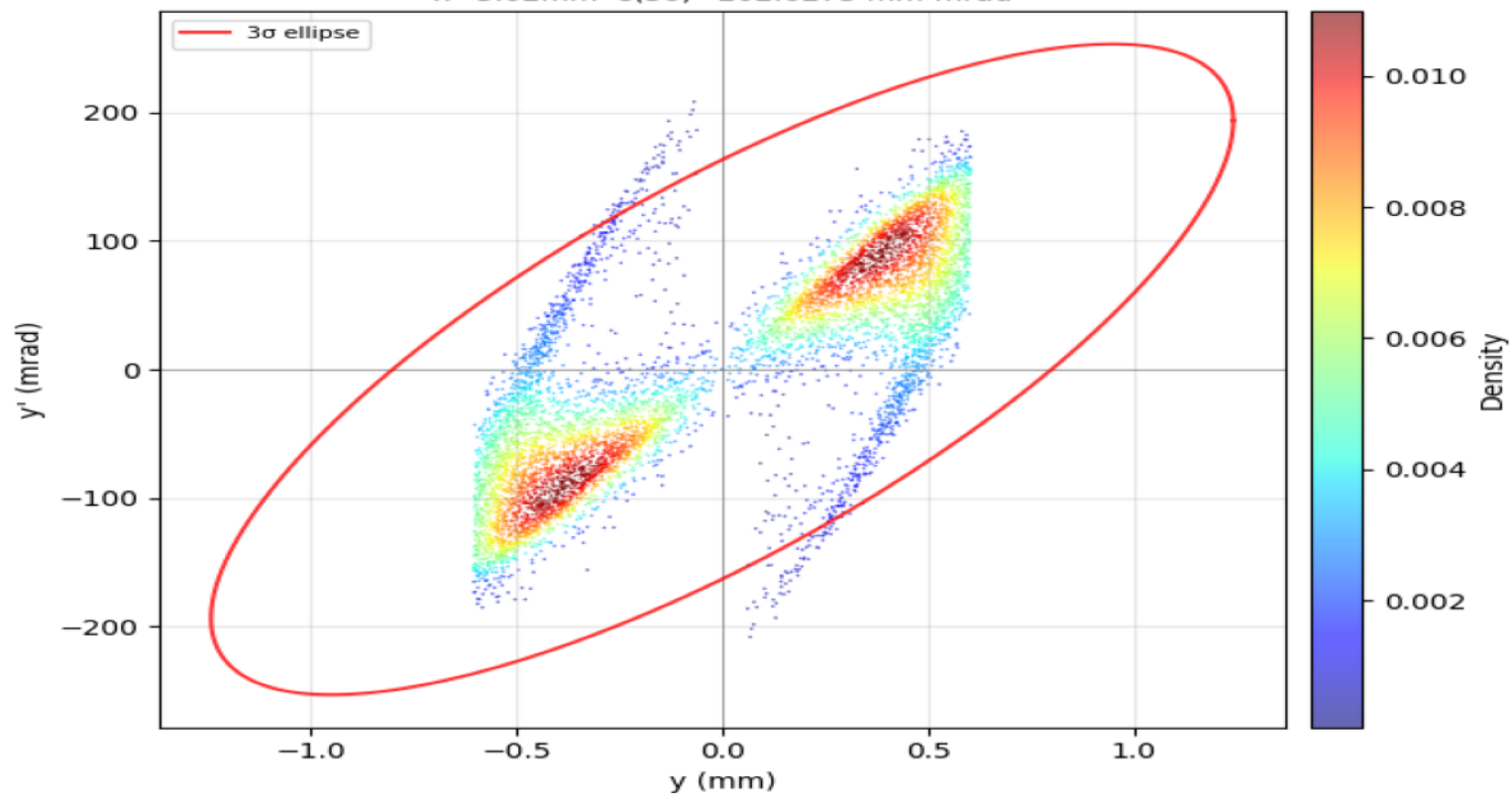
--- Results at Electrode Exit ---
Exit position: 3.00 mm
Divergence: 76.552 mrad (4.386°)
RMS radius: 0.4218 mm
Beam current: 3.1675e-04 A
Grid interception: 8.0%
Transmission: 92.0%
```

Particle Trajectories (Axisymmetric)

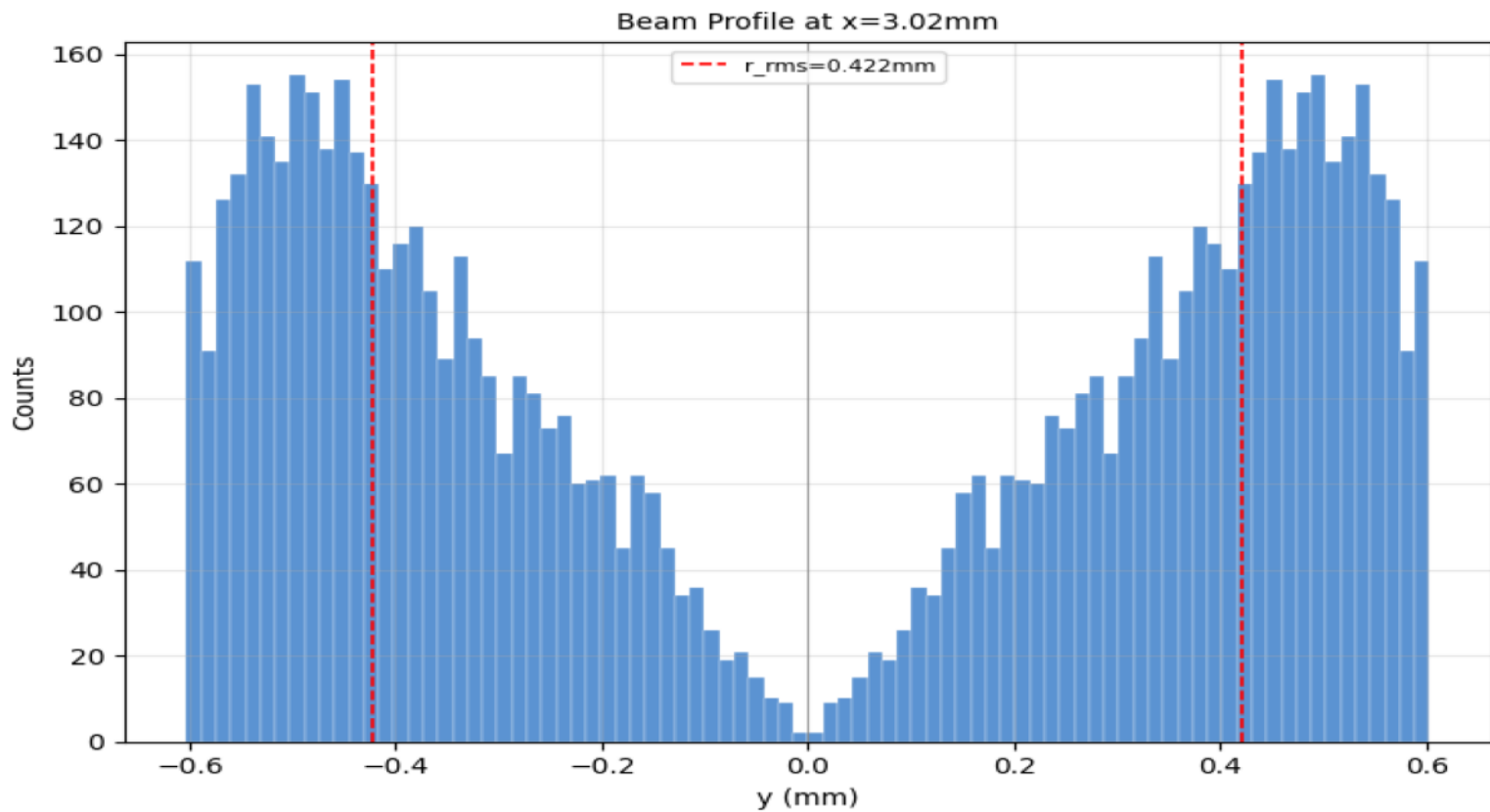


# Phase Space

$x=3.02\text{mm}$   $\epsilon(3\sigma)=202.6273\text{ mm-mrad}$

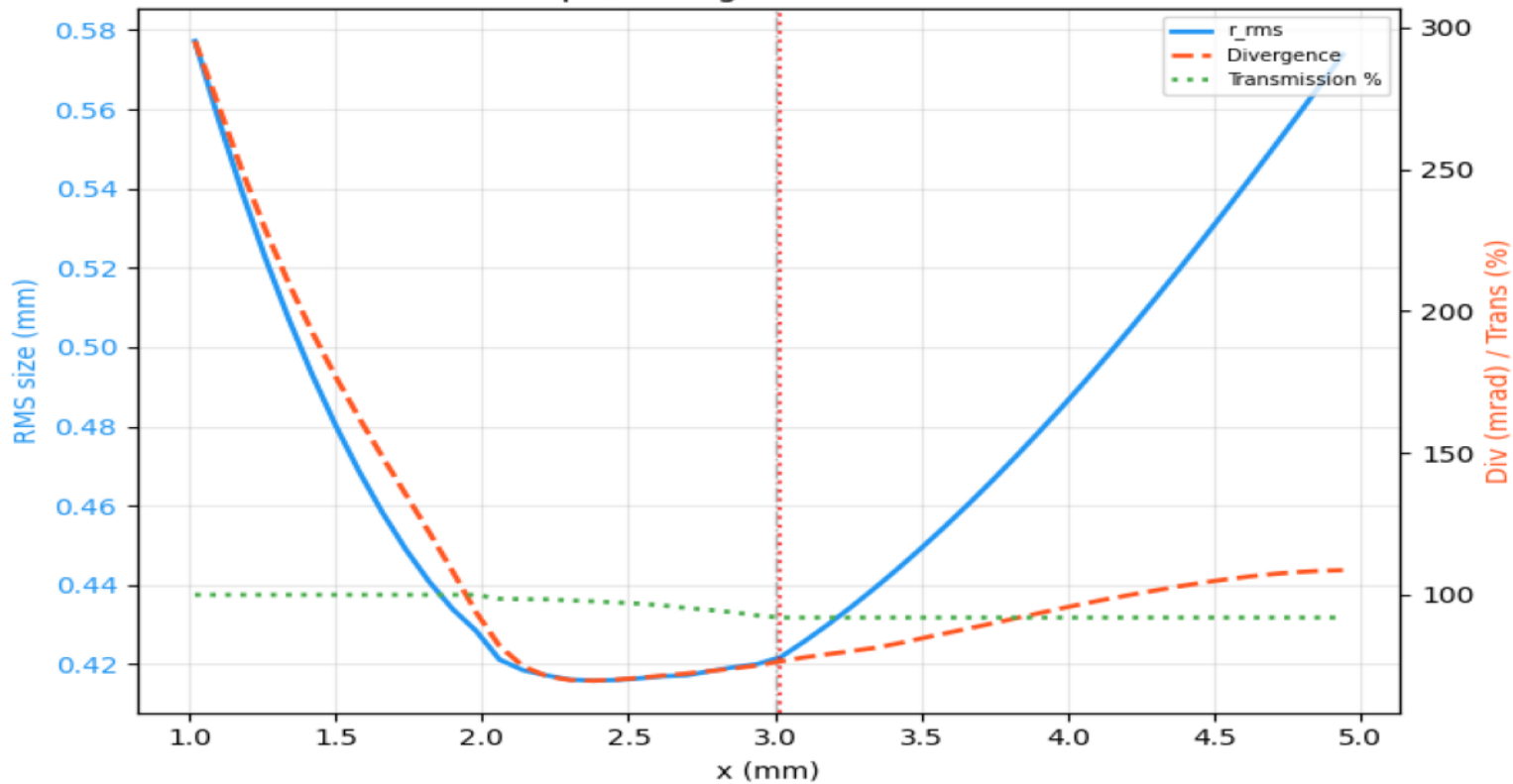


# Beam Profile



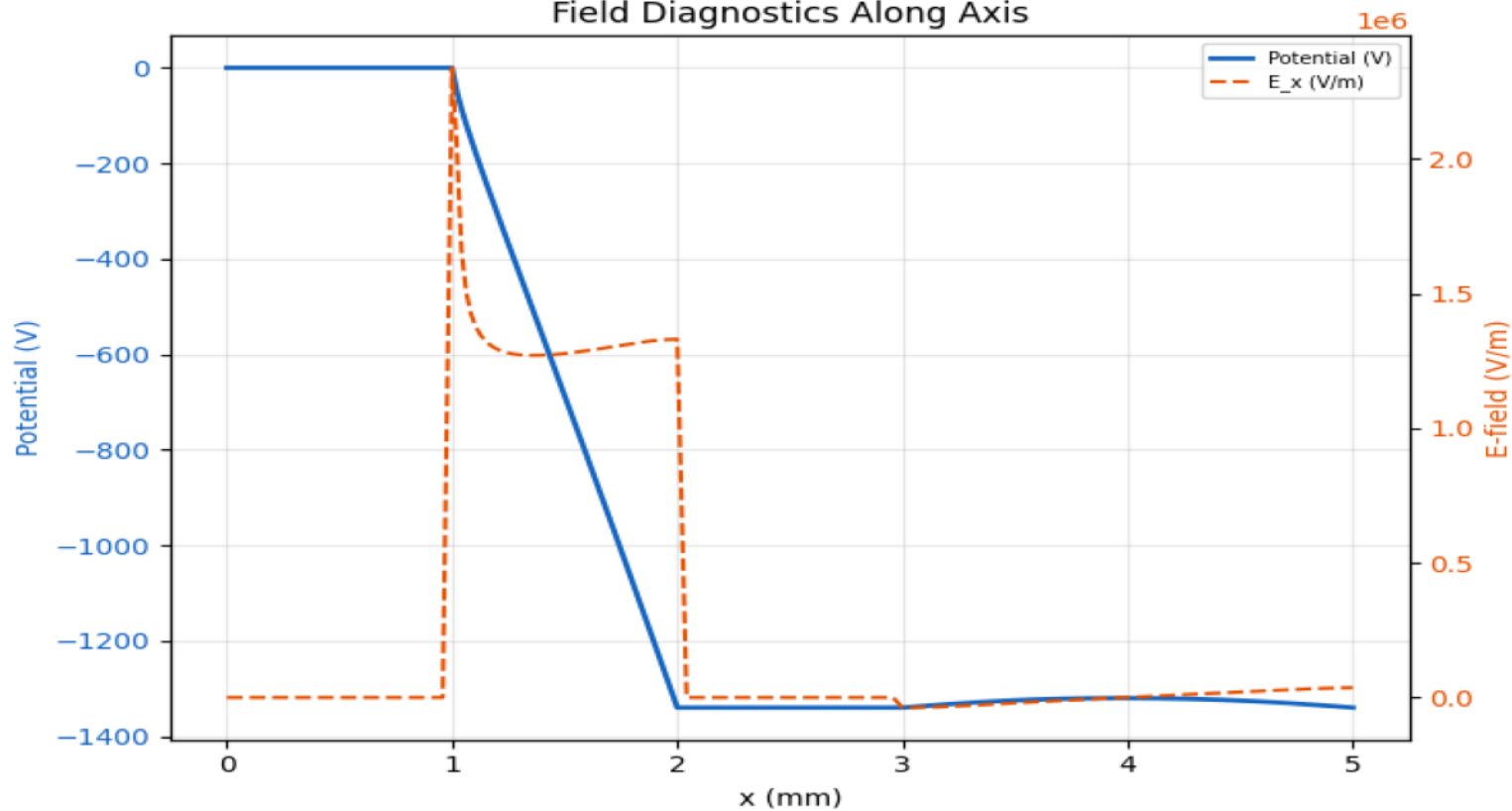
# Envelope, Divergence & Transmission

## Envelope, Divergence & Transmission



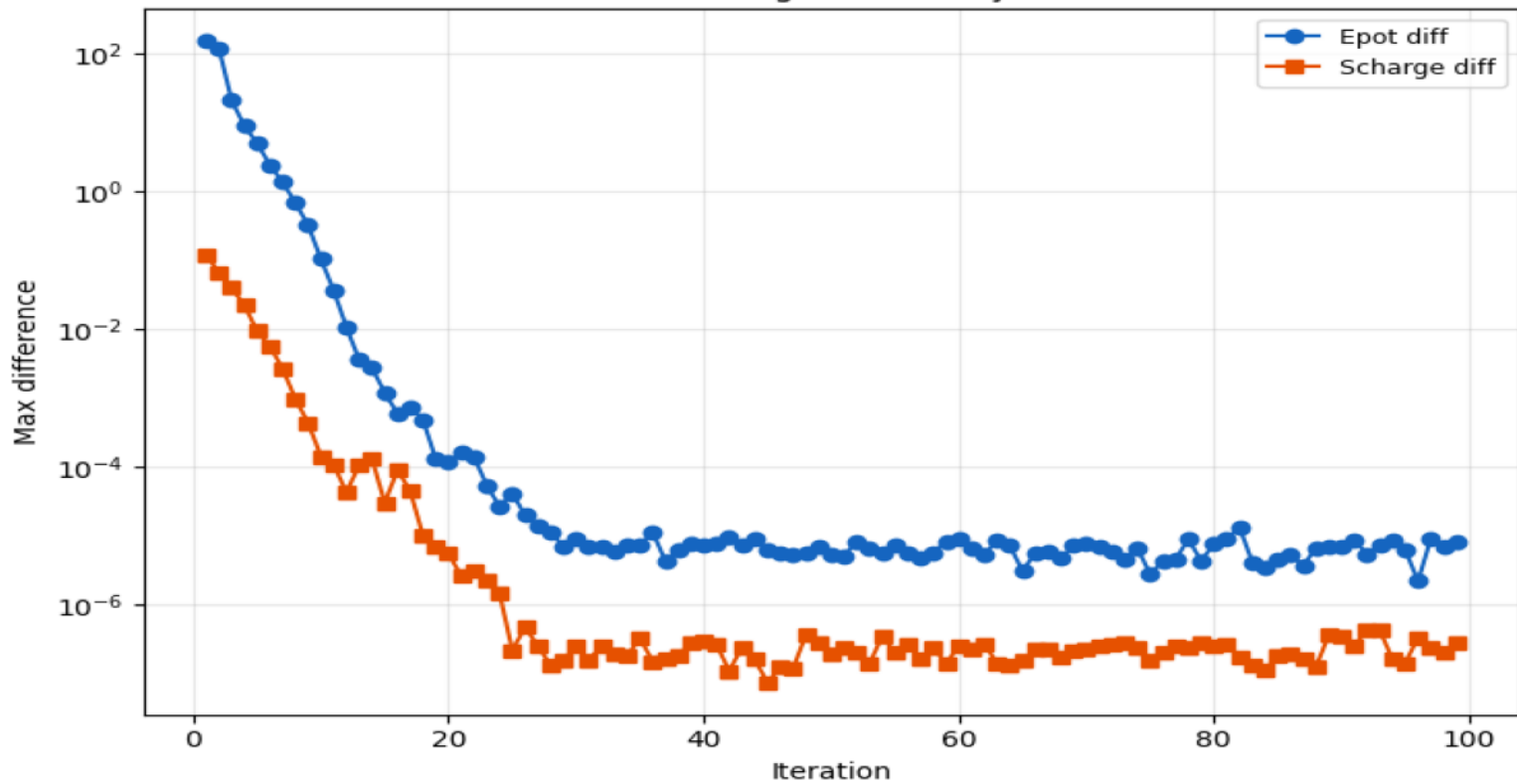
# Field Diagnostics

## Field Diagnostics Along Axis



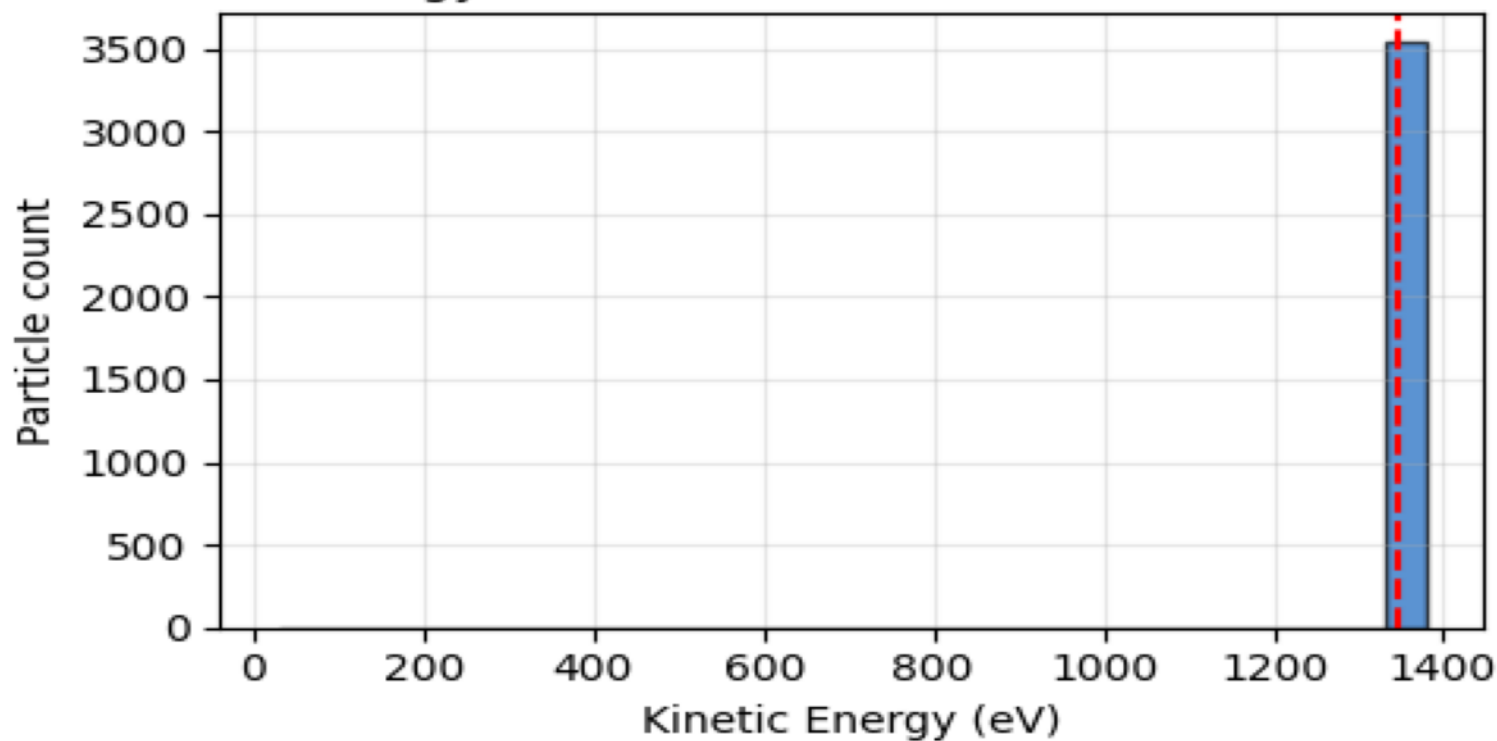
# Convergence

Convergence History



# Energy Distribution

Energy at x=4.9mm  $E=1345.3\pm22.29$  eV



# Perveance Scan

## Perveance Scan (Current)

